Date: Roll No.

III SEMESTER B.Tech (IT, ITNS)

MID-SEMESTER EXAMINATION 2025

Course Code-INECC305/ITECC305 Time: 1:30 Hrs

Course Title-Probability & Stochastic Processes Max Marks: 20

Attempt all questions. Missing data / information (If any) may be suitably assumed & mentioned in the answer.

Q.No.	Questions	Marks	CO
1	a) Let the events A and B have $P[A]=x$, $P[B]=y$, and $P[A\cup B]=z$. Find $P[A\cap B]$, $P[A^c\cap B^c]$, $P[A^c\cup B^c]$, $P[A\cap B^c]$ and $P[A^c\cup B]$.	03	CO1
	b) Find $P[A\|B]$ if $A\cap B=\emptyset$; if $A\subset B$; if $A\supset B$.	01	CO2
2	a) A target is made of three concentric circles of radii $3^{-\frac{1}{2}}$, 1 and $3^{1/2}$ feet. Shots within the inner circle count 4 points, within the next ring 3 points, and within the third ring 2 points. Shots outside of the target count 0. Let R be the random variable representing distance of the hit from the centre. Suppose that the PDF of R is $$f_R(r)=\begin{cases}\dfrac{2}{\pi(1+r^2)} & \text{for } r>0\\ 0 & \text{elsewhere}\end{cases}$$ Compute the probability of scoring 0, 2, 3 & 4 points.	03	CO1
	b) Determine which of the following are valid distribution functions: i. $F_X(x)=\begin{cases}1-e^{-\frac{x}{2}} & x\ge 0\\ 0 & x<0\end{cases}$ ii. $F_X(x)=\begin{cases}0 & x<0\\ 0.5+0.5\sin\left[\frac{\pi(x-1)}{2}\right] & 0\le x<2\\ 1 & x\ge 2\end{cases}$ iii. $F_X(x)=\frac{x}{a}[u(x-a)-u(x-2a)]$, here u(.) is the unit step function	01	CO1
3	a) A random variable X has the distribution function $$F_X(x)=\sum_{n=1}^{12}\frac{n^2}{650}u(x-n)$$ Find the probabilities: (i) $P\{-\infty<X\le 6.5\}$ (ii) $P\{X>4\}$, and (iii) $P\{6<X\le 9\}$	03	CO2

	b) A voltage X is uniformly distributed in the set$\{-3,\ldots\ldots,3,4\}$. Find the mean and variance of $Y=-2X^2+3$.	01	CO2
4	a) A random variable X is uniformly distributed on the interval (-5, 15). Another random variable $Y=e^{-\frac{X}{5}}$ is formed. Find the probability density function of Y. Also compute $E[Y]$ and $E[Y^2]$.	03	CO2
	b) Express $P[\|Y\|>\alpha,\|X\|>\beta]$ in terms of the joint CDF of X & Y.	01	CO2
5	a) Let X and Y have joint pdf: $f_{X,Y}(x,y)=k(x+y)$ $\quad for\ 0\le x\le 1, 0\le y\le 1$ i. Find k, ii. Find the marginal pdf of X and of Y. iii. Find $P[X<Y]$, $P[Y<X^2]$, $P[X+Y>0.5]$.	03	CO2
	b) Define the correlation coefficient 'ρ' between two random variables X & Y. Interpret the values of correlation coefficient	01	CO2

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